Project GyerAR: The GeciKAzan Protocol for Low-Cost Augmented Reality and Textile-Based Body-Tracking Infrastructure

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Abstract

This document outlines the theoretical framework and implementation specifications for an ultra-low-cost, decentralized augmented reality (AR) headset coupled with a modular, textile-integrated whole-body motion capture suite. Designated as the **GeciKAzan Architecture**, this platform relies strictly on *First Principles* to strip commercial wearable ecosystems down to their bare-metal physical and algorithmic fundamentals. By bypassing centralized infrastructure, proprietary waveguide optoelectronics, and external lighthouse tracking systems, the complete architecture delivers real-time sub-15ms spatial virtualization for a bill of materials (BOM) cost under 38,000 HUF (∼110 USD).

1 Introduction and First Principles

Modern spatial computing configurations (e.g., Apple Vision Pro, Microsoft HoloLens) introduce extreme mechanical and financial overhead via complex diffractive optical elements, heavy application-specific integrated circuits (ASICs), and active active-matrix time-of-flight sensors.

By applying a *First Principles* deconstruction to coordinate extraction and heads-up telemetry, the **GeciKAzan Protocol** establishes that whole-body AR interfaces require only four fundamental vectors:

1. **Ambient Environmental Ingestion:** An untethered, low-latency image sensor to digitalize the forward field of view.
2. **Decentralized Kinematic Sampling:** Independent, low-power processing nodes capable of calculating rotational vectors and joint flexions locally.
3. **Non-Obstructive Retinal Emission:** A transparent optical combiner that projects digital coordinates onto the user's eye without diminishing ambient real-world photons.
4. **Autonomous Energy Capture:** A safe, lightweight electrochemical power distribution matrix optimized to mitigate organic tissue degradation.

2 Unified System Bill of Materials (BOM)

The component distribution and maximum cost ceilings for the complete integrated network (AR Glasses + 5-Node Body Capture Matrix) are structured in Table 1.

Table 1: GeciKAzan Ecosystem Hardware Budget (HUF / USD)

Subsystem	Component / Description	Price (HUF)	Price (USD)
AR Headset	ESP32-CAM (with OV2640 Sensor)	2,500 – 3,500	\$8 – \$10
	0.96" / 1.3" I2C Monochrome OLED	1,500 – 2,500	\$4 – \$7
	Structural Frame + Polymer Combiner	1,000 – 3,000	\$3 – \$9
Trackers (x5)	ESP32-PICO-D4 / S3 Mini Node SoCs	10,000 – 15,000	\$28 – \$42
	MPU6050 / BNO055 6/9-Axis IMUs	5,000 – 12,500	\$14 – \$35
	Carbon-Infused Conductive Rubber Cords	4,000 – 7,500	\$11 – \$21
Power Matrix	Li-Po Cells + 5V Inductive Qi Coils	4,000 – 8,000	\$11 – \$22
TOTAL	**Full Interactive AR Array**	**28,000 – 38,000**	**\$80 – \$110**

3 Kinematic and Optical Mechanics

3.1 Joint Flexion via Conductive Materials

Instead of relying on multi-joint optical computer vision tracking, finger and extremity articulation are mapped via textile-integrated carbon-infused conductive rubber cords. The base electrical resistance (R_{rubber}) shifts linearly under structural tension (ΔL). By integrating the conductive polymer into a fixed-resistance voltage divider circuit, the raw elongation is converted directly into readable voltage differentials for the microcontroller's Analog-to-Digital Converter (ADC):

$$V_{\text{out}} = V_{\text{in}} \cdot \left(\frac{R_{\text{rubber}}}{R_{\text{fix}} + R_{\text{rubber}}} \right) \tag{1}$$

Spatial orientation for each extremity is simultaneously extracted via 6-axis Inertial Measurement Units (IMUs) passing raw gyroscope and accelerometer matrices through an on-chip Madgwick orientation filter to resolve stable Euler/Quaternion orientation states.

3.2 The Pepper's Ghost Reflection and Coordinate Transformation

Heads-up informational transparency is resolved by mounting the emissive monochrome OLED screen face-down on the brow rim of the frame, projecting light downwards onto a transparent rigid polymer sheet (cut from a standard poly-carbonate case) set at a strict 45° angle.

Because the physical optics yield an inverted mirrored plane, the bare-metal C code executes an instantaneous horizontal coordinate transformation directly within the SSD1306 display driver registers to avoid software memory overhead:

$$I_{\text{render}}(x, y) = I_{\text{buffer}}(W - 1 - x, y) \tag{2}$$

where W is the screen width in pixels.

4 Power Architecture: Electromagnetic Induction Matrix

Traditional far-field Over-the-Air (OTA) radiofrequency (RF) power beaming was analyzed and rejected due to strict physical limits. First, the inverse-square law governs radiative power transfer efficiency:

$$P_{\text{rx}} \propto \frac{1}{d^2} \tag{3}$$

resulting in a power efficiency drop below 0.1% across joint distances. Second, the high water content of human muscle tissue creates a severe dielectric loss barrier at 2.4 GHz – 5.8 GHz, inducing localized tissue heating.

The **GeciKAzan Core** bypasses this safely via a passive textile-integrated inductive array. Flat copper receiver coils are sewn directly into the inner garment layers. When garments are dropped onto a centralized magnetic induction pad matrix, all nodes establish local magnetic coupling to safely fast-charge at 5W per tracking zone without physical port wear.

5 Distributed Software Cryptographic Game Engine

The network pipeline operates completely decoupled via Bluetooth Low Energy (BLE 5.0 GATT) and local UDP packets, delivering compressed 15-byte tracking data frames at 90 Hz. The processing layer is handled via a host Python engine running light YOLOv8 object identification models.

To maintain cryptographic verification and protocol resilience, hidden localized alternate-reality ciphers (The Satoshi Games) are hardcoded directly into the tracking loop. These sequences dynamically test real-time positional changes against cryptographic keys, validating true peer-to-peer system synchronization without central server authorizations.

6 Conclusion

The **GeciKAzan Architecture** proves that whole-body immersion and spatial augmentation can be successfully deployed at a fraction of standard commercial costs. By replacing heavy computing infrastructure with highly optimized C algorithms, clever optical arrangements, and raw physics, the system presents an indestructible blueprint for street-level hacker autonomy and open engineering exploration.